AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
990 A	226.30 MVA	89.5 degC	81.2 degC	39.354 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		1200	mm2
Conductor type		milliken	
Conductor diameter Dc		43.1	mm
Conductor screen OD Dc_sc		46.7	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		80.5	mm
Ins outer screen OD Di_sc		82.9	mm
Insulation thickness t_ins		16.10	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
Screen type		Cu_wire	
Screen mean diameter ds		84.3	mm
Screen area As		140.2	mm2
Armour type		none	
Oversheath material		PE	
Cable outer diameter De		98.0	mm
Rated voltage U		132.0	kV
U0 line-to-earth		76.210	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		98.0	mm
Ambient ground temp theta_a		20	degC
Soil thermal resistivity rho_soil		1.00	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		1.5100e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 20 degC		1.5100e-05	ohm/m
R0 at theta_max = 90 degC		1.9254e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant (IEC 60287-1-1 Table 2)		0.8000	
kp proximity constant (IEC 60287-1-1 Table 2)		0.3700	
xs^2 = ks*(8pif/R0)*1e-7		5.2213001299	
xs■ = $(xs^2)^2$		27.2619750469	
ys skin effect factor		0.1275058631	
xp■		5.8315068499	
yp" intermediate proximity		0.0296519501	
yp proximity factor		0.0229311323	
R at theta_max AC = R0(1+ys+yp)		2.2150525415e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
U0 line-to-earth		76.2100	kV
Di insulation outer diameter		80.5000	mm
Dc_sc conductor screen OD		46.7000	mm
Capacitance C = er/(18 ln(Di/Dc_sc)) 10^-9		0.255070	nF/m
omega = 2 pi f		314.159265	rad/s
Wd = omega C U0^2 tan(delta)		4.6540712851e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		140.2500	mm2
Wire screen outer OD D_sc_outer		87.8400	mm
Lay factor LFs		1.136863	
Screen Rs0 at 20 degC (with LFs)		1.3976e-04	ohm/m
--- Al laminated foil ---			
Foil thickness t_fl		0.2000	mm
Foil outer diameter D_fl		88.8000	mm
Foil area A_fl		55.669022	mm2
Foil Rso_fl at 20 degC		5.0772e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins insulation		3.50	K.m/W
rho_sc semiconducting screens		2.50	K.m/W
rho_tape bedding tape under wire screen		12.00	K.m/W
T1_cs conductor screen		0.0319189884	K.m/W
T1_i insulation		0.3033167853	K.m/W
T1_si insulation screen		0.0116890861	K.m/W
T1_uwbt bedding tape under wire screen		0.0319840375	K.m/W
T1_raw = sum of all layers		0.3789088974	K.m/W
T1 correction x1.00 (foil=True, cover=42.4%)		0.3789088974	K.m/W
D_sc_outer = D_tuws + 2*ds_wire (for T2)		87.8400	mm
D_owt tape over wire screen (for T2)		88.4000	mm
T2 screen to foil tape		0.0121371405	K.m/W
T3_raw oversheath (two-layer)		0.0532813189	K.m/W
T3 x1.6 touching trefoil factor		0.0852501103	K.m/W
u = 2L/De		20.408163	
T4 soil		1.4701534385	K.m/W
Sum T = T1+T2+T3+T4		1.9464495866	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		81.2113	degC
Rs at theta_s		1.3610e-04	ohm/m
X sheath reactance		5.0784e-05	ohm/m
lam1' circulating current		0.750955	
lam1'' eddy current		0.041020	
lam1 = lam1' + lam1''		0.791976	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		70.00	K
Etop numerator		69.182282	
Ebot denominator		7.0614e-05	
I = sqrt(Etop/Ebot) exact		989.8122	A
I rounded TB880 GP2		990	A
MVA = sqrt(3) x U x I		226.301	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		89.5224	degC
theta_s sheath surface		81.2113	degC
theta_j cable outer surface		77.8564	degC
theta_a ambient ground		20.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I^2 x R		21.701496	W/m
Sheath Ws = lam1 x Wc		17.187059	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		4.6541e-01	W/m
Total		39.353962	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.168747	uH/m
Reactance X = omega L		53.0133	uOhm/m
Pos-seq resistance R1		39.6932	uOhm/m
Pos-seq impedance Z1		66.2266	uOhm/m
Capacitance C		0.255070	nF/m
Charging current Ic		6.1069	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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